25900235 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Value Addition Course (VAC) - MG3VACBCH200 - MICROPLASTICS AND ENVIRONMENT

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer all questions

Each question carries **1** marks

1. Microfibers are types of [U] / [CO1]

   a). Nanoplastics b). Natural fibers

   c). Macroplastics d). Microplastics

2. Which of the following summarizes the human health risk of microplastic biomagnification? [U] / [CO1]

   a). Causes improved digestion b). Potential endocrine disruption, inflammation, and toxicity

   c). Enhances immunity d). Boosts calcium absorption

3. The effect of microplastic in plants can cause. [U] / [CO2]

   a). Unlimited flowering cycles. b). Increase in leaf area index in all conditions

   c). Permanent immunity to drought d). Impaired root growth and function

4. Which uncertainty is biological in nature? [K] / [CO3]

a). Unpredictable interactions with body

b). Unknown recycling rates

c). Limited polymer classification

d). Known dose-response

5. [K] / [CO4]

What happens when polar sea ice melts in relation to microplastics?

a). Microplastics are destroyed

b). Microplastics are released into the oceans

c). Greenhouse gases are absorbed

d). Microbial communities stop functioning

[1x5 = 5]

Part B

Fill In The Blank Questions

Answer all questions

Each question carries **2** marks

6. The growing concern about microplastics is mainly due to their potential impacts on environmental and _____. [U] / [CO1]

7. Microbeads are plastic particles smaller than _____ millimeters. [K] / [CO1]

8. Microplastics negatively impact the _____ industry by reducing seafood safety and availability. [K] / [CO2]

9. Microplastics can reduce _____ quality in males, affecting fertility. [K] / [CO3]

10. In adsorption method _____and _____are used as adsorbents for removing microplastics. [U] / [CO4]

[2x5 = 10]

Part C

Short Answer Questions

Answer any **5** questions

Each question carries **3** marks

11. Explain how rivers act as pathways in transporting microplastics to oceans [U] / [CO1]

12. Define bioaccumulation. Name one heavy metal that undergoes bioaccumulation in aquatic organisms. [K] / [CO1]

13. Discuss the significance of classifying microplastics into different categories. [U] / [CO1]

14. List any three key effects of microplastics on freshwater ecosystems. [U] / [CO2]

15. Recall the sources of microplastics in drinking water. [K] / [CO3]

16. Mention the areas where safety limits for microplastics are set by regulatory agencies? [K] / [CO4]

17. Explain how microplastics cause habitat disruption and alter food webs in marine ecosystems. [U] / [CO3]

[3x5 = 15]

Part D

Short Essay Questions

Answer any **4** questions

Each question carries **5** marks

18. Demonstrate the possible mechanisms to minimise the human exposure to microplastics [U] / [CO1]

19. Assess the effectiveness of chemical versus biological methods for microplastic degradation, highlighting their advantages and limitations. [E] / [CO1]

20. Evaluate the different types of physiological, behavioral, and ecological responses exhibited by wildlife to microplastic exposure. [E] / [CO2]

21. Give the impact of different types of pollutants on microplastics distribution in food. [U] / [CO3]

22. Discuss how microplastics can alter soil structure and fertility. [U] / [CO1]

23. Evaluate the impact of microplastics on the cryosphere, microbial and biogeochemical processes, focusing on zooplankton and carbon cycling in marine ecosystems. [E] / [CO4]

[5x4 = 20]